

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.

3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.

2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Question 1

Given

- Network: **192.168.100.0/24**
- School of Computing → **/25**
- School of Electronics → **/26**
- School of Mechanical Engineering → **/27**

School of Computing (/25)

- **Network Address:** 192.168.100.0
- **Host Range:** 192.168.100.1 – 192.168.100.126
- **Broadcast Address:** 192.168.100.127
- **Usable Hosts:** 126

School of Electronics (/26)

- **Network Address:** 192.168.100.128
- **Host Range:** 192.168.100.129 – 192.168.100.190
- **Broadcast Address:** 192.168.100.191
- **Usable Hosts:** 62

School of Mechanical Engineering (/27)

- **Network Address:** 192.168.100.192
- **Host Range:** 192.168.100.193 – 192.168.100.222
- **Broadcast Address:** 192.168.100.223
- **Usable Hosts:** 30

Remaining Unused Range

192.168.100.224 – 192.168.100.255

	Department	Network	Host Range	Broadcast	Usable Hosts
	Computing	192.168.100.0/25	192.168.100.1–126	192.168.100.127	126
	Electronics	192.168.100.128/26	192.168.100.129–190	192.168.100.191	62
	Mechanical	192.168.100.192/27	192.168.100.193–222	192.168.100.223	30

Conclusion: The network is successfully divided into three subnets, and the remaining addresses are available for future expansion.

Question 2

Given

- Network: **10.10.0.0/16**
- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

	Department	Network Address	Host Range	Broadcast	Usable Hosts
	Software Development	10.10.0.0/24	10.10.0.1–254	10.10.0.255	254
	Quality Assurance	10.10.1.0/25	10.10.1.1–126	10.10.1.127	126
	Human Resources	10.10.1.128/26	10.10.1.129–190	10.10.1.191	62

Department	Network Address	Host Range	Broadcast	Usable Hosts
Executive Management	10.10.1.192/27	10.10.1.193–222	10.10.1.223	30

Remaining Unused Range

10.10.1.224 – 10.10.255.255

Conclusion: Each department receives a subnet according to its required size, leaving the remaining address space for future use.

Question 3

Given IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000/64

Compressed Address

2001:db8:abcd::/64

Expanded Address

2001:0db8:abcd:0000:0000:0000:0000:0000

Network Prefix

2001:db8:abcd:0::/64

Number of Host Addresses

Host bits = **64**

Total host addresses:

$2^{64} = 18,446,744,073,709,551,616$

Item	Answer
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Compressed Address	2001:db8:abcd::/64
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Expanded Address	2001:0db8:abcd:0000:0000:0000:0000:0000
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Network Prefix	2001:db8:abcd:0::/64
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Item	Answer
Host Addresses	18,446,744,073,709,551,616

Conclusion: The IPv6 address is compressed using "::", and a /64 network provides 2^{64} possible host addresses.

Question 4

Given Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: **192.168.2.45**

Answer

- **Destination Subnet:** 192.168.2.0/24
- **Matching Routing Entry:** 192.168.2.0/24
- **Next-Hop Interface:** Interface connected to the 192.168.2.0/24 network
- **Default Route:** 0.0.0.0/0 (used if no route matches)

Task	Answer
Destination Subnet	192.168.2.0/24
Matching Route	192.168.2.0/24
Next-Hop	Interface connected to 192.168.2.0/24
Default Route	0.0.0.0/0

Conclusion: The router forwards the packet through the interface connected to the **192.168.2.0/24** network. If no matching route exists, it uses the default route.

Question 5

Given

- Administration LAN → **192.168.10.0/26**
- Finance LAN → **192.168.10.64/26**

Administration LAN

- **Network Address:** 192.168.10.0
- **Host Range:** 192.168.10.1 – 192.168.10.62
- **Broadcast Address:** 192.168.10.63
- **Default Gateway:** 192.168.10.1

Suggested PCs:

- 192.168.10.2
- 192.168.10.3
- 192.168.10.4

Finance LAN

- **Network Address:** 192.168.10.64
- **Host Range:** 192.168.10.65 – 192.168.10.126
- **Broadcast Address:** 192.168.10.127
- **Default Gateway:** 192.168.10.65

Suggested PCs:

- 192.168.10.66
- 192.168.10.67
- 192.168.10.68

LAN	Network	Host Range	Broadcast	Gateway
Administration	192.168.10.0/26	192.168.10.1–62	192.168.10.63	192.168.10.1

LAN	Network	Host Range	Broadcast	Gateway
Finance	192.168.10.64/26	192.168.10.65 – 126	192.168.10.127	192.168.10.65

Conclusion: Both LANs are connected through a Layer-3 router. Hosts use their respective default gateways to communicate within and outside their subnet.